

Project Report

Project : Constant g_m Bias

4-Transistor OTA via Negative Feedback

Course	Analog Integrated Circuit Design
Department	Electrical Engineering
Institute	Indian Institute of Technology Kanpur
Tool	Cadence Virtuoso (Spectre Simulator)
Technology	CMOS (1.8 V supply)

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Introduction

Transconductance (g_m) is one of the most critical parameters in analog circuit design. It directly governs the gain, bandwidth, and noise performance of amplifiers, filters, and data-converters. However, g_m is inherently sensitive to process, voltage, and temperature (PVT) variations. A transistor biased with a fixed current will show significant g_m drift because $g_m = \sqrt{2\mu_n C_{ox}(W/L)I_D}$ for an NMOS in saturation, where mobility μ_n decreases with temperature.

This project addresses the problem by designing a **constant- g_m bias circuit** using a compact **4-transistor OTA (VCCS)** architecture governed by a negative feedback servo loop. The feedback loop continuously senses the g_m of a replica OTA and adjusts the bias such that

$$G_m = \frac{I_{\text{ref}}}{2\Delta V} \quad (1)$$

where I_{ref} is a stable current reference and $2\Delta V$ is a fixed incremental voltage generated from a resistive voltage divider referenced to V_{DD} .

Design Specifications

Table 1: Project Performance Targets

Parameter	Target	Constraint
Architecture	4-transistor OTA (VCCS)	–
Target g_m	10 mS	At nominal PVT
Control Mechanism	Circuit-based negative FB	Self-regulating
g_m Variation	< 1%	–20 °C to +100 °C
Supply Voltage (V_{DD})	1.8 V	–

Working Principle

Negative Feedback Servo Loop

The core idea is borrowed from a well-known servo-loop technique for g_m setting. A small incremental DC voltage $2\Delta V$ is applied differentially to the input of a replica 4T OTA.

The resulting output current $2G_m\Delta V$ is compared against a stable reference current I_{ref} using an error amplifier G_{err} .

The error amplifier drives the common-mode bias voltage V_{cm} of the OTA. If G_m increases (due to temperature drop or process variation), the output current exceeds I_{ref} , causing V_{cm} to decrease, which in turn reduces G_m — a classic negative feedback action. At steady state:

$$2G_m\Delta V = I_{\text{ref}} \implies G_m = \frac{I_{\text{ref}}}{2\Delta V} \quad (2)$$

Since I_{ref} and ΔV are both derived from stable references, G_m is locked to a PVT-insensitive value.

Voltage Divider for $2\Delta V$

The incremental voltage is generated from a symmetric resistive divider:

$$2\Delta V = V_{\text{ref}} \frac{R_1}{R + R_1} \quad (3)$$

In the implementation, $R_0 = R_3 = 8.989 \text{ k}\Omega$ and $R_1 = R_2 = 100 \Omega$, giving a small ΔV that keeps the OTA well within its linear region.

Common-Mode Feedback

A common-mode feedback (CMFB) path sets the output common-mode voltage $V_{op,om}$ to V_{cm} . The two op-amp blocks (I4 and I5) serve as error amplifiers: one (G_{err}) drives the g_m servo loop, while the other enforces the output common-mode level via a feedback path.

Circuit Architecture

Top-Level Negative Feedback Loop (`negative_fb_loop`)

The top-level schematic implements the complete servo system:

- **BLOCK** – The 4-transistor OTA replica whose g_m is being regulated.
- **R0, R3** (8.989 k Ω) – Upper/lower arms of the voltage divider.
- **R1, R2** (100 Ω) – Middle taps that define $\pm\Delta V$.

- **R4, R5** ($1\text{ M}\Omega$) – Averaging resistors for V_{op} and V_{om} .
- **I1, I2** ($94\text{ }\mu\text{A}$) – Reference current sources (I_{ref}).
- **I4, I5** – Error amplifiers (Opamp instances) closing the servo and CMFB loops.
- **C1** (2 pF) – Compensation capacitor for loop stability.

4-Transistor OTA (BLOCK)

The BLOCK is a differential-pair OTA with a current-mirror load:

- **PM0, PM1** (PMOS, $W = 6\text{ }\mu\text{m}$, $L = 540\text{ nm}$, $m = 24$) – Current mirror load.
- **NM0, NM1** (NMOS, $W = 9\text{ }\mu\text{m}$, $L = 540\text{ nm}$, $m = 14$) – Differential pair.
- **R0** ($100\text{ }\Omega$) – Source degeneration resistor to linearise G_m .
- Ports: `Vin_Plus`, `Vin_minus`, `Vcm_fb`, `Vom`, `Vop`.

Error Amplifier (Opamp – I4 / I5)

Each error amplifier is a standard 5-transistor OTA:

- **PM0, PM1** (PMOS, $W = 2\text{ }\mu\text{m}$, $L = 2\text{ }\mu\text{m}$, $m = 11$) – Current mirror load.
- **NM0, NM1** (NMOS, $W = 2\text{ }\mu\text{m}$, $L = 2\text{ }\mu\text{m}$, $m = 11$) – Differential pair.
- **I0** ($20\text{ }\mu\text{A}$) – Tail current source.
- Ports: `Vin_plus`, `Vin_minus`, `Vout`.

Operating Point Analysis

All operating points were extracted using Cadence Spectre DC analysis at nominal temperature (27°C) and $V_{DD} = 1.8\text{ V}$.

4T OTA (BLOCK) – Transistor Operating Points

Table 2: DC Operating Points – 4T OTA (BLOCK)

headerblue				
rowgray Role	Mirror Load	Mirror Load	Diff. Pair	Diff. Pair
Region	Saturation	Saturation	Saturation	Saturation
rowgray V_{GS} (mV)	−831.1	−831.1	+758.9	+739.1
V_{DS} (mV)	−796.9	−884.1	+792.5	+705.3
rowgray V_{TH} (mV)	−464.2	−464.2	+560.0	+560.0
V_{DSAT} (mV)	−324.1	−324.1	+204.8	+190.4
rowgray I_{DS} (μ A)	−1050.1	−1056.2	+1144.0	+962.2
g_m (mS)	5.028	5.058	9.249	8.480
rowgray g_{ds} (μ S)	73.38	66.43	73.30	69.94
g_m/I_D	4.789	4.789	8.084	8.813
rowgray r_{out} (k Ω)	13.63	15.05	13.64	14.30
Self Gain ($g_m r_{out}$)	68.5	76.1	126.2	121.2

Observations:

- All four transistors are firmly in **saturation**, as required for proper OTA operation.
- The NMOS differential pair (NM0/NM1) exhibits higher g_m (≈ 8.5 – 9.2 mS) than the PMOS mirror load (≈ 5.0 mS), consistent with the higher electron mobility.
- The g_m/I_D ratio for NMOS (≈ 8 – 8.8) indicates moderate inversion, while PMOS (≈ 4.8) operates at a higher overdrive, which is expected for a current mirror with wide transistors.
- Source degeneration ($R_0 = 100\ \Omega$) linearises the overall G_m at the cost of a slight reduction in effective transconductance.

Error Amplifier I5 – Transistor Operating Points

Table 3: DC Operating Points – Error Amplifier I5 (Opamp)

headerblue				
rowgray Role	Mirror Load	Mirror Load	Diff. Pair	Diff. Pair
Region	Saturation	Saturation	Saturation	Saturation
rowgray V_{GS} (mV)	−598.6	−598.6	+582.4	+583.3
V_{DS} (mV)	−598.6	−831.1	+824.3	+591.8
rowgray V_{TH} (mV)	−437.3	−437.2	+559.4	+559.5
V_{DSAT} (mV)	−167.8	−167.8	+97.4	+97.9
rowgray I_{DS} (μ A)	−9.97	−10.03	+9.97	+10.03
g_m (μ S)	98.81	99.30	174.3	175.1
rowgray g_{ds} (nS)	281.2	247.2	351.9	386.3
g_m/I_D	9.91	9.90	17.48	17.46
rowgray r_{out} (M Ω)	3.56	4.05	2.84	2.59
Self Gain	351.4	401.7	495.3	453.3

Error Amplifier I4 – Transistor Operating Points

Table 4: DC Operating Points – Error Amplifier I4 (Opamp)

headerblue				
rowgray Role	Mirror Load	Mirror Load	Diff. Pair	Output/Tail
Region	Saturation	Saturation	Saturation	Saturation*
rowgray V_{GS} (mV)	−656.4	−656.4	+615.8	+528.7
V_{DS} (mV)	−656.4	−839.7	+756.3	+573.1
rowgray V_{TH} (mV)	−437.2	−437.2	+561.9	+562.0
V_{DSAT} (mV)	−211.0	−211.1	+116.7	+71.6
rowgray I_{DS} (μ A)	−16.60	−16.67	+16.60	+3.40
g_m (μ S)	129.7	130.2	259.9	68.9
rowgray g_{ds} (nS)	414.3	364.7	534.8	155.7
g_m/I_D	7.81	7.81	15.66	20.27
rowgray r_{out} (M Ω)	2.41	2.74	1.87	6.42
Self Gain	313.1	356.9	486.0	442.4

*NM1 in I4 operates at region code 3 (near saturation/subthreshold boundary) due to its low $I_{DS} = 3.4 \mu\text{A}$ compared to the main pair.

Observations on Amplifiers I4 and I5:

- Both amplifiers are biased at low currents ($\approx 10\text{--}17 \mu\text{A}$ per branch), resulting in high output resistances in the M Ω range, which translates to large DC gain well suited for the feedback loop.
- The very high self-gain values (300–495) indicate that both op-amps provide sufficient loop gain to enforce $G_m = I_{\text{ref}}/2\Delta V$ accurately.
- NMOS transistors in both amplifiers operate in moderate inversion ($g_m/I_D \approx 15\text{--}20$), prioritising gain efficiency over speed.
- V_{DSAT} values are small ($< 220 \text{ mV}$), ensuring adequate output swing headroom within the 1.8 V supply.

Simulation Results and Specification Status

g_m vs. Temperature — Observed Result

A DC sweep simulation was performed over temperature from -20°C to $+100^\circ\text{C}$ using Cadence Spectre. The transconductance was probed directly from the device operating point of NM0 (OS("/I0/NM0" "gm")) in the closed-loop `negative_fb_loop` schematic.

Table 5: Simulated g_m vs. temperature — actual results

Temp ($^\circ\text{C}$)	g_m (mS)	Deviation from 27°C	Spec ($< 1\%$)
-20	9.80	$+5.95\%$	FAIL
0	9.55	$+3.24\%$	FAIL
27	9.25	Reference	—
50	8.90	-3.78%	FAIL
75	8.55	-7.57%	FAIL
100	8.20	-11.35%	FAIL

Result: The g_m varies from $\approx 9.80\text{ mS}$ at -20°C to $\approx 8.20\text{ mS}$ at $+100^\circ\text{C}$ — a total variation of $\sim 19\%$, far exceeding the target of $< 1\%$. The design **does not meet** the specification.

Discussion — Why the $< 1\%$ Specification is Not Met

The dominant reason the specification is not achieved is the **temperature dependence of carrier mobility**. In a MOSFET biased at a fixed drain current, the transconductance is:

$$g_m = \sqrt{2 \mu_n(T) C_{ox} (W/L) I_{DS}}$$

where $\mu_n(T) \propto T^{-3/2}$ decreases with rising temperature. Since the bias current I_{DS} in the 4T OTA is set by the tail source — which itself has a finite temperature coefficient in a real implementation — g_m falls nearly linearly with temperature, producing the observed $\sim 19\%$ drift.

The negative feedback servo loop is intended to counteract this by increasing I_{DS} (via raising V_{cm_fb}) as temperature rises, so that the product $\mu_n(T) \cdot I_{DS}(T)$ remains constant. However, achieving this requires the loop to track the mobility variation *precisely* and in *real time*, which places a stringent demand on the loop gain and the accuracy of I_{ref} and

ΔV .

In this implementation, the available loop correction is insufficient to fully compensate the steep mobility roll-off across the -20°C to $+100^\circ\text{C}$ range. Additionally, polysilicon resistors used to generate ΔV carry a non-negligible temperature coefficient ($\approx +1500\text{ ppm}/^\circ\text{C}$), which causes ΔV — and hence the target locking point $G_m = I_{\text{ref}}/2\Delta V$ — to shift slightly with temperature, further degrading the regulation accuracy. These two effects together account for the observed gap between the simulated performance and the $< 1\%$ target.

Common-Mode Voltages (net6/net8) vs. Temperature

The feedback node voltages (net6, net8) vary in the range 965–980 mV across temperature, showing a non-monotonic U-shaped profile. Ideally, if the servo loop were functioning correctly, V_{cm_fb} should sweep significantly (by several hundred mV) to actively compensate the falling g_m . The narrow variation observed ($\sim 15\text{ mV}$) strongly suggests the loop is not exerting meaningful control over the bias point.

Design Calculations

Correct g_m Locking Point

For the servo to lock $G_m = 10\text{ mS}$, the required $2\Delta V$ with $I_{\text{ref}} = 94\text{ }\mu\text{A}$ is:

$$2\Delta V_{\text{required}} = \frac{I_{\text{ref}}}{G_m} = \frac{94\text{ }\mu\text{A}}{10\text{ mS}} = 9.4\text{ mV} \quad (4)$$

The current resistor values produce:

$$2\Delta V_{\text{actual}} = 1.8\text{ V} \times \frac{100}{9089} \approx 19.8\text{ mV} \quad \Rightarrow \quad G_{m,\text{locked}} \approx 4.75\text{ mS} \quad (5)$$

To correct this without changing I_{ref} , R_1 should be reduced:

$$R_1 = \frac{2\Delta V_{\text{required}}}{V_{DD} - 2\Delta V_{\text{required}}} \times R = \frac{9.4\text{ mV}}{1.8\text{ V} - 9.4\text{ mV}} \times 8989\text{ }\Omega \approx 47\text{ }\Omega \quad (6)$$

Alternatively, I_{ref} can be increased to $198\text{ }\mu\text{A}$ keeping the existing resistor values.

Effective G_m with Source Degeneration

With source degeneration resistor $R_S = 100\ \Omega$ (split: $50\ \Omega$ per side), the effective OTA transconductance is:

$$G_{m,\text{eff}} = \frac{g_m}{1 + g_m (R_S/2)} = \frac{9.25\ \text{mS}}{1 + 9.25\ \text{mS} \times 50\ \Omega} \approx 8.13\ \text{mS} \quad (7)$$

The servo loop must account for this degeneration when locking G_m to the target.

Output Resistance of 4T OTA

$$R_{\text{out}} = r_{o,\text{NM1}} \parallel r_{o,\text{PM1}} = 14.30\ \text{k}\Omega \parallel 15.05\ \text{k}\Omega \approx 7.33\ \text{k}\Omega \quad (8)$$

Required Loop Gain for $< 1\%$ Regulation

To suppress the inherent 19% open-loop variation down to $< 1\%$:

$$\left. \frac{\Delta G_m}{G_m} \right|_{\text{CL}} = \frac{\Delta G_m / G_m|_{\text{OL}}}{1 + T} < 1\% \quad \Rightarrow \quad 1 + T > \frac{19\%}{1\%} = 19 \quad \Rightarrow \quad T > 18 \quad (9)$$

With self-gain of NM0 ≈ 126 and the error amplifier gain > 300 , the available loop gain is theoretically sufficient. The issue lies in how effectively this gain is utilised in the actual feedback path.

Summary and Conclusion

This project involved the design and simulation of a constant- g_m bias circuit using a compact 4-transistor OTA with a negative feedback servo loop in Cadence Virtuoso Spectre.

What Was Achieved

- A 4T OTA with NMOS differential pair and PMOS mirror load was successfully designed with all transistors biased in **saturation** at nominal conditions ($V_{DD} = 1.8\ \text{V}$, 27°C).
- Two high-gain error amplifiers (I4, I5) with individual transistor self-gain > 300 were designed and verified through operating point extraction.
- The complete servo loop schematic — including voltage divider, reference current sources, CMFB path, and compensation capacitor — was implemented in Virtuoso.

- Temperature sweep simulation from -20°C to $+100^\circ\text{C}$ was performed and the g_m characteristics were extracted and analysed.

What Was Not Achieved and Why

- The $< 1\%$ g_m variation specification was **not met**. The simulated device-level g_m varies by $\approx 19\%$ across the temperature range.
- The root causes identified are: (a) a mismatch between I_{ref} and $2\Delta V$ that sets the wrong locking point; (b) the servo loop may not be actively regulating the bias — evidenced by the near-constant V_{cm_fb} across temperature; (c) the probed quantity ($g_{m,NM0}$) is the device-level parameter, not the closed-loop G_m regulated by the servo.
- These are **correctable issues** and do not indicate a fundamental flaw in the circuit topology, which is consistent with the reference design.

Learning Outcomes

The project provided hands-on experience in: hierarchical schematic design in Cadence Virtuoso, transistor-level operating point analysis using Spectre, understanding PVT sensitivity of g_m and the role of negative feedback in suppressing it, and systematic debugging of analog feedback loops — including identifying loop connectivity issues, gain-bandwidth tradeoffs, and measurement methodology errors.

The fundamental circuit topology is sound and consistent with the reference design. With further tuning of the resistive divider, a bandgap-referenced I_{ref} , and loop stability verification via AC analysis, the circuit is expected to achieve the $< 1\%$ regulation target.

References

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